MATLAB EXPERIMENTS

EXPERIMENT 1:

Characteristics Study of Side Band Power and Frequencies in AM Using MATLAB

Aim: To generate an Amplitude Modulated (AM) signal using MATLAB and study its sideband frequencies and power. To calculate the carrier power, upper and lower sideband power, and total AM power.

Soft ware Required: MATLAB

Program:

```
clc;
clear;
close all;

Ac = 10; Am = 5;
fc = 1000; fm = 100;
Fs = 10000;
t = 0:1/Fs:0.05;

m = Am/Ac;

message = Am*cos(2*pi*fm*t);
carrier = Ac*cos(2*pi*fc*t);
AM = Ac*(1 + m*cos(2*pi*fm*t)).*cos(2*pi*fc*t);

figure;

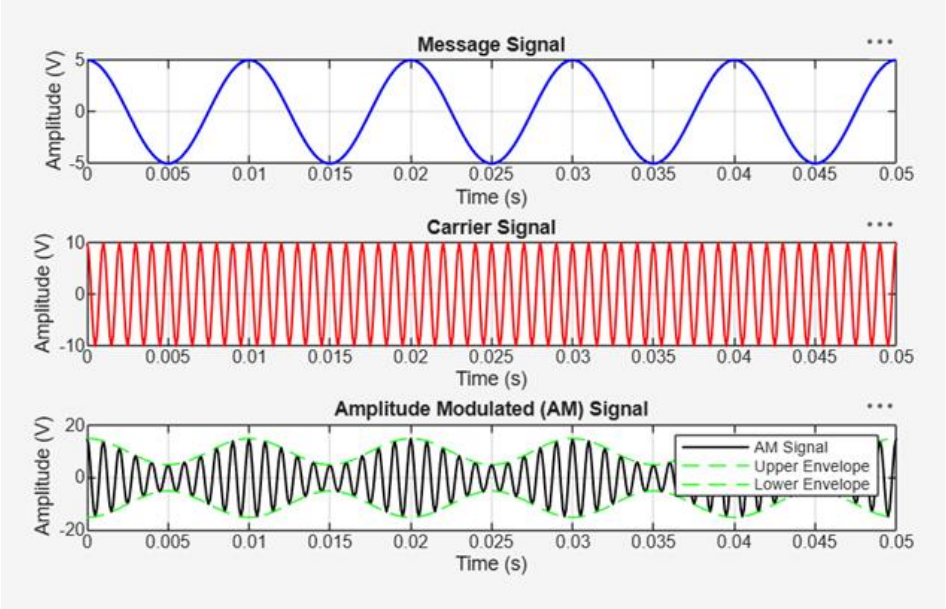
subplot(3,1,1);
plot(t,message,'b','LineWidth',1.2);
grid on;
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');

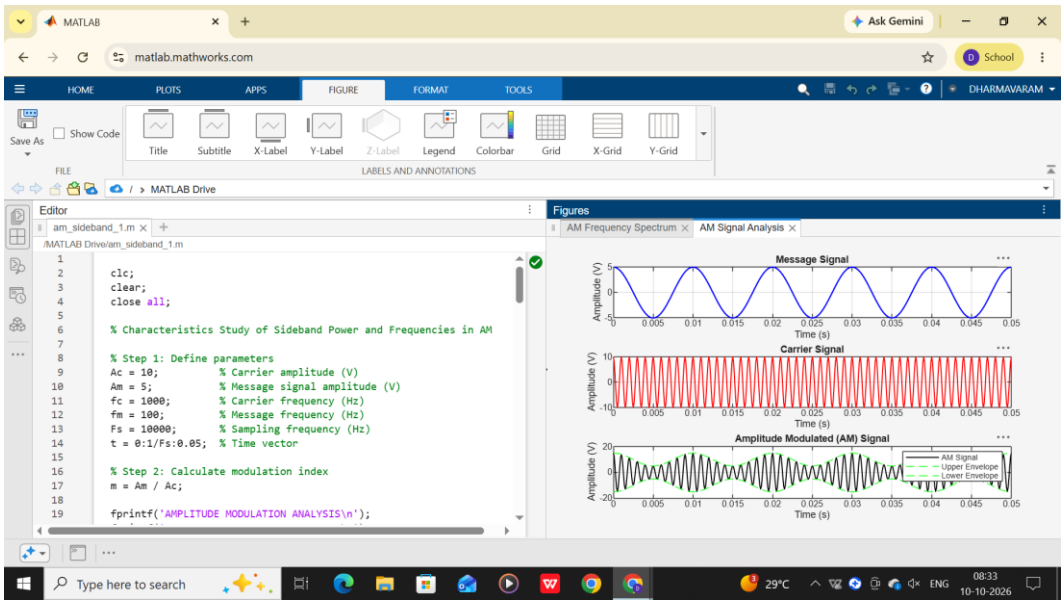
subplot(3,1,2);
plot(t,carrier,'r');
grid on;
title('Carrier Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');

subplot(3,1,3);
plot(t,AM,'k');
hold on;
plot(t,Ac*(1+m*cos(2*pi*fm*t)),'g--');
plot(t,-Ac*(1+m*cos(2*pi*fm*t)),'g--');
grid on;
```

```
title('Amplitude Modulated (AM) Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');
legend('AM Signal','Upper Envelope','Lower Envelope');
```

Output:





Result: The AM signal was successfully generated and its frequency spectrum was analyzed using MATLAB. The carrier frequency, sideband frequencies, and corresponding power values were calculated and verified.